INCH-POUND

MIL-S-22885/111(EC)
AMENDMENT 1
10 May 1996

MILITARY SPECIFICATION SHEET

SWITCHES, PUSHBUTTON, LIGHT EMITTING DIODE (LED) ILLUMINATED, STANDARD, NIGHT VISION IMAGING SYSTEM (NVIS) AND SUNLIGHT READABLE DISPLAYS, SPDT AND DPDT, LOW LEVEL TO 7 AMPERES, SEALED AND UNSEALED, SOLDER AND COMMON TERMINATION SYSTEM (CTS) TERMINATIONS

This amendment forms a part of MIL-S-22885/111, dated 14 August 1992, and is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

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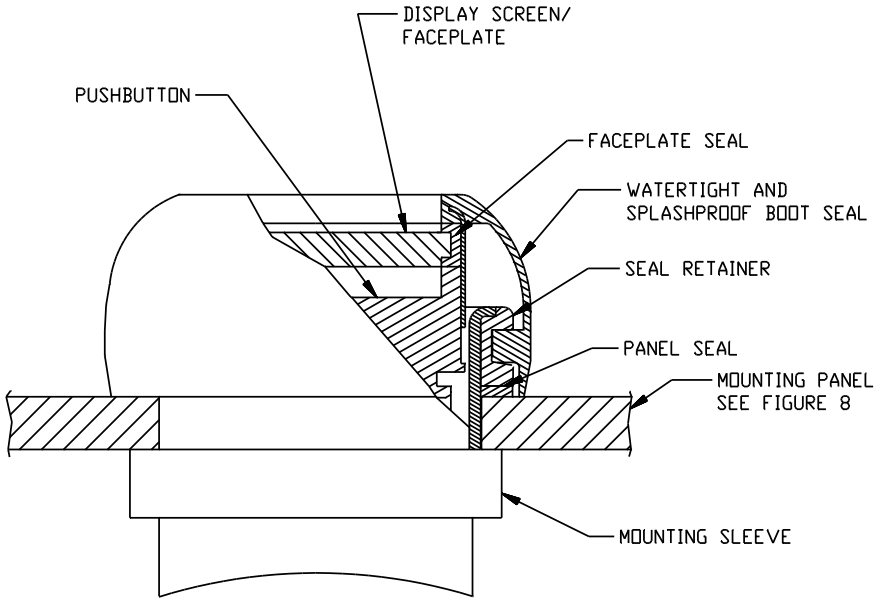
FIGURE 1, side view, delete dimension ".20".

PAGE 2

FIGURE 2: top view, delete dimension ".90" and substitute ".98 MAX"; side view, delete dimension ".43" and substitute ".39"; seal cross sectional view, delete and substitute the following:



SEAL CROSS SECTIONAL VIEW

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FIGURE 3: side view, delete dimension ".20"; side view, delete dimension "1.21" and substitute "1.59"; side view, delete "MIL-C-39029/57-357" and substitute "M39029/16-168"; NOTE 7, delete "M39029/57-357" and substitute "M39029/16-168".

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FIGURE 4: top view, delete dimension ".90" and substitute ".98 MAX"; side view, delete ".43" and substitute ".39"; side view, delete "MIL-C-39029/57-357" and substitute "M39029/16-168"; seal cross sectional view, delete and substitute the same view as shown for figure 2 above; NOTE 7, delete "M39029/57-357" and substitute "M39029/16-168".

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FIGURE 9: socket contact, crimp type, delete "MIL-C-39029/57-357" and substitute "M39029/16-168"; mounting cam, delete "RECEPTACEL ASSYEMBLY" and substitute "RECEPTACLE ASSEMBLY".

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Operating characteristics:

Actuating force, delete "Unsealed: 2 pounds ±1 pound", "Dripproof: 3 pounds ±1 pound", "Splashproof/watertight: 4 pounds ±1 pound" and substitute "5 pounds maximum".

Delete "Actuating travel: .125 inch maximum, 0.08 inch minimum" and substitute "Plunger travel: .130 ± .030".

Pushbutton removal force, delete "2 to 4 pounds" and substitute "2 to 5 pounds. Not applicable to type III or type VI (sealed) switches."

Shock, delete and substitute the following:

"Shock:

All switch types: Method I (75 g).

Switch types III and VI (sealed switches): High impact shock in accordance with MIL-S-901, grade A, class II (for deck-mounted equipment) when attached to the anvil plate of the shock machine by means of standard mounting fixture 11-C with simulated console, which provides resilient mounting typical of in-service use within "deck-mounted" electronic cabinets or panels. These units are not recommended for direct "hull mounted" applications."

Salt spray, delete and substitute the following:

"Salt spray: Test condition A (96 hours)."

After sand and dust paragraph, add the following new paragraph:

"Terminal Strength:

Solder terminals, applied force and direction of force: 3 pounds perpendicular to the long axis, and 5 pounds parallel to the long axis."

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TABLE I, delete and substitute the following:

" TABLE I. Illuminated color and luminance limits.

Color	Color code	Nominal peak wave length (nm)	Color limits 1/		Minimum average luminance 2/ (100% duty cycle)
			x	y	
Red	RD	630	.668 .682 .705 .719	.304 .318 .267 .281	60 foot-Lamberts
Amber yellow	AY	585	.465 .479 .589 .603	.506 .520 .382 .396	70 foot-Lamberts
Lime green	LG	565	.324 .338 .499 .513	.644 .659 .472 .485	150 foot-Lamberts
Green	GR	555	.252 .266 .395 .409	.710 .724 .575 .589	45 foot-Lamberts
Orange	OR	610	.613 .627 .678 .692	.358 .372 .294 .308	70 foot-Lamberts
Ultra red	UR	660	.700 .714 .713 .733	.272 .286 .258 .267	400 foot-Lamberts
NVIS green B	JJ	553	In accordance with MIL-L-85762 3/		40 foot-Lamberts
NVIS yellow	KK	580			50 foot-Lamberts
NVIS red	LL	607			8 foot-Lamberts

1/ Corners of the x and y envelope limits from the CIE chromaticity diagram.
2/ With or without EMI/RFI shielding.
3/ Appropriate power modulation required to regulate LED illumination output."

NVIS compatibility, NVIS red, first line, delete "type II class A".

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Marking, add the following: "Pushbuttons shall indicate the applicable nominal voltage (5v or 28v). LED printed circuit boards shall be blue for current sinking configurations and natural or green for current sourcing configurations. NVIS type and class marking not required. The word "TOP" shall be marked on switch housings and pushbuttons, excluding splashproof/watertight sealed units, to indicate proper orientation."

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Qualification, group submission, at the end of the paragraph add "Captive pushbutton retainers are not required on assemblies used for qualification and retention testing."

TABLE X, delete and substitute the following:

" TABLE X. Qualification inspection - group submission.

Test samples	Qualification inspection table of MIL-S-22885		Extent of approval
	Group	Number of samples	
M22885/111B213D	I II 1/ III VI 2/ VII	10 2 from group I 2 from group I 2 from group I 4 from group I	All PINs
M22885/111A812A	I 3/ II 1/ VII 4/	4 2 from group I 2 from group I	
M22885/111Y241K	I II 5/ 6/ III V 6/ VI 7/ VII 6/ 8/	12 4 from group I 2 from group I 2 from group I 2 from group I 2 from group I	
M22885/111B213D	VIII 9/	42 10/	

1/ Shock: Method I.
2/ Electrical endurance: Sea level, inductive dc load only.
3/ Test current and voltage for contact resistance and operating characteristics tests shall not exceed 10 mA and 30 mV dc on all gold plated contact switches subject to low level life test. Low level life contact resistance requirements apply.
4/ Mechanical endurance test not required.
5/ Shock: High impact shock in accordance with MIL-S-901, grade A, class II.

TABLE X. Qualification inspection - group submission - continued.

6/ Sealing shall be verified by the watertight and splashproof tests only.
7/ Electrical endurance: Sea level, resistive dc load only.
8/ Intermediate current test not required.

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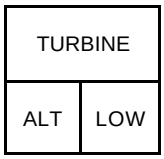
9/ NVIS color and radiance measurements shall be made, as applicable, during group VIII inspections.

10/ In combination with the test sample specified in the first column, 42 pushbutton display screen assemblies shall be tested: Two each of six colors for each standard display type, N and W; two each of two colors for display type S; and 2 each of three NVIS colors and four standard colors for display type H. One of the pushbutton samples for each color/display type combination shall have 5 volt LED's, and the other shall have 28 volt LED's. Test legend characters and display screen design shall be as follows:



"

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APPLICATION INFORMATION, after U.L. listing, add the following: "Recommended substitute for color white is LG. Best color separation is obtained with combinations of LG, OR, and UR; or with GR, AY, and RD."

APPLICATION INFORMATION, Oil, fuel, and solvent resistance, delete in its entirety.

Category II product offerings, add the following:

"Display screens with four-way splits are available.
Oil, fuel, and solvent resistant seals are available (A listing of specific types of chemicals, recommended, and not recommended, may be obtained from the switch manufacturer.)"

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TABLE XII, 3rd column, minimum lighted contrast ratios: Orange, delete ".25" and substitute ".40"; Amber yellow, delete ".20" and substitute ".30"; Red, delete ".20" and substitute ".25"; Green, delete ".20" and substitute ".30".

CONCLUDING MATERIAL

Custodians:
Navy - EC

Preparing activity:
DLA - ES

(Project 5930-1625)

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